

Contrastive Divergence

Derivation from KL Divergence and Energy-Based Models

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April 30, 2026

These notes derive contrastive divergence (CD) from the Kullback–Leibler divergence and connect it to:

- 1 classical Boltzmann machines (BMs),
- 2 general energy-based models (EBMs),
- 3 restricted Boltzmann machines (RBMs),

The central question is:

Why is the approximate BM gradient called *contrastive divergence*?

The Learning Problem

Let $p_{\text{data}}(x)$ be the data distribution and $p_{\theta}(x)$ a parametric model.

Goal: find θ such that $p_{\theta} \approx p_{\text{data}}$.

Maximum likelihood (ML) maximises

$$\mathcal{L}(\theta) = \mathbb{E}_{p_{\text{data}}}[\log p_{\theta}(x)] = \sum_x p_{\text{data}}(x) \log p_{\theta}(x).$$

Equivalence with KL minimisation.

$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = \sum_x p_{\text{data}}(x) \log \frac{p_{\text{data}}(x)}{p_{\theta}(x)} = \underbrace{H(p_{\text{data}})}_{\text{const.}} - \mathcal{L}(\theta),$$

so minimising $D_{\text{KL}}(p_{\text{data}} \| p_{\theta})$ is exactly equivalent to maximising $\mathcal{L}(\theta)$.

KL Divergence: Key Properties

- $D_{\text{KL}}(p\|q) \geq 0$, with equality iff $p = q$ (Gibbs inequality).
- D_{KL} is *not* symmetric: $D_{\text{KL}}(p\|q) \neq D_{\text{KL}}(q\|p)$ in general.
- $D_{\text{KL}}(p\|q)$ penalises regions where $p > 0$ but $q \approx 0$ (mode-covering behaviour for ML estimation).

Proof of non-negativity (Jensen). Since $-\log$ is convex:

$$D_{\text{KL}}(p\|q) = \mathbb{E}_p \left[-\log \frac{q(x)}{p(x)} \right] \geq -\log \mathbb{E}_p \left[\frac{q(x)}{p(x)} \right] = -\log \sum_x q(x) = 0.$$

An **energy-based model** assigns an energy $E_\theta(x) \in \mathbb{R}$ to each configuration x :

$$p_\theta(x) = \frac{e^{-E_\theta(x)}}{Z_\theta}, \quad Z_\theta = \sum_x e^{-E_\theta(x)}.$$

Taking the log:

$$\log p_\theta(x) = -E_\theta(x) - \log Z_\theta.$$

The fundamental difficulty: Z_θ is a sum over all configurations, which is exponentially large for high-dimensional x .

Log-Likelihood Gradient: Full Derivation

$$\nabla_{\theta} \mathcal{L}(\theta) = \mathbb{E}_{p_{\text{data}}} [\nabla_{\theta} \log p_{\theta}(x)] = -\mathbb{E}_{p_{\text{data}}} [\nabla_{\theta} E_{\theta}(x)] - \nabla_{\theta} \log Z_{\theta}.$$

Gradient of the partition function.

$$\begin{aligned} \nabla_{\theta} \log Z_{\theta} &= \frac{\nabla_{\theta} Z_{\theta}}{Z_{\theta}} = \frac{\nabla_{\theta} \sum_x e^{-E_{\theta}(x)}}{Z_{\theta}} \\ &= \frac{-\sum_x e^{-E_{\theta}(x)} \nabla_{\theta} E_{\theta}(x)}{Z_{\theta}} = -\mathbb{E}_{p_{\theta}} [\nabla_{\theta} E_{\theta}(x)]. \end{aligned}$$

Substituting back:

$$\boxed{\nabla_{\theta} \mathcal{L} = -\mathbb{E}_{p_{\text{data}}} [\nabla_{\theta} E_{\theta}] + \mathbb{E}_{p_{\theta}} [\nabla_{\theta} E_{\theta}].}$$

Positive and Negative Phases

For *minimising* the negative log-likelihood $-\mathcal{L}$:

$$\nabla_{\theta}(-\mathcal{L}) = \underbrace{\langle \nabla_{\theta} E_{\theta} \rangle_{\text{data}}}_{\text{positive phase}} - \underbrace{\langle \nabla_{\theta} E_{\theta} \rangle_{\text{model}}}_{\text{negative phase}}.$$

Positive phase Lowers E_{θ} on observed data configurations. Estimated cheaply from the training set.

Negative phase Raises E_{θ} on configurations preferred by the current model. Requires samples from p_{θ} — generally intractable.

The negative phase is the computational bottleneck of all EBM training.

Classical Boltzmann Machine

A **Boltzmann machine** has visible units $v = (v_i)$ and hidden units $h = (h_j)$, with energy

$$E_{\theta}(v, h) = - \sum_i a_i v_i - \sum_j b_j h_j - \sum_{ij} W_{ij} v_i h_j - \sum_{i < k} U_{ik} v_i v_k - \sum_{j < \ell} V_{j\ell} h_j h_{\ell}.$$

The joint and visible distributions are

$$p_{\theta}(v, h) = \frac{e^{-E_{\theta}(v, h)}}{Z_{\theta}}, \quad p_{\theta}(v) = \sum_h p_{\theta}(v, h).$$

The log-likelihood gradient over visible data is

$$\nabla_{\theta} \mathcal{L} = -\mathbb{E}_{p_{\text{data}}(v)} [\mathbb{E}_{p(h|v)} [\nabla_{\theta} E_{\theta}(v, h)]] + \mathbb{E}_{p_{\theta}(v, h)} [\nabla_{\theta} E_{\theta}(v, h)].$$

Restricted Boltzmann Machine: Architecture

In an **RBM** visible-visible and hidden-hidden connections are removed:

$$E_{\theta}(v, h) = -a^{\top} v - b^{\top} h - v^{\top} W h.$$

This bipartite structure gives *conditional independence*:

$$p_{\theta}(h \mid v) = \prod_j p_{\theta}(h_j \mid v), \quad p_{\theta}(v \mid h) = \prod_i p_{\theta}(v_i \mid h).$$

Binary units. For $v_i, h_j \in \{0, 1\}$:

$$p(h_j = 1 \mid v) = \sigma(b_j + \sum_i W_{ij} v_i),$$

$$p(v_i = 1 \mid h) = \sigma(a_i + \sum_j W_{ij} h_j),$$

where $\sigma(x) = 1/(1 + e^{-x})$.

Deriving the Conditional Probabilities

From the energy,

$$p(h_j = 1 \mid v) = \frac{p_\theta(v, h_j = 1)}{p_\theta(v, h_j = 0) + p_\theta(v, h_j = 1)}.$$

The numerator's energy contribution from h_j is $-(b_j + \sum_i W_{ij} v_i) h_j$. Therefore

$$\frac{e^{b_j + \sum_i W_{ij} v_i}}{1 + e^{b_j + \sum_i W_{ij} v_i}} = \sigma(b_j + \sum_i W_{ij} v_i).$$

Because v - h are bipartite, the h_j are *conditionally independent* given v , so $p_\theta(h \mid v) = \prod_j p_\theta(h_j \mid v)$. This makes block Gibbs sampling exact and efficient.

Since $\partial E / \partial W_{ij} = -v_i h_j$, the gradient of the log-likelihood is

$$\frac{\partial \mathcal{L}}{\partial W_{ij}} = \langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_{\text{model}}.$$

Estimating the positive phase (easy):

$$\langle v_i h_j \rangle_{\text{data}} = \mathbb{E}_{p_{\text{data}}(v)} [v_i \sigma(b_j + W_{:j}^\top v)].$$

Estimating the negative phase (hard): requires equilibrium Gibbs samples

$$\{v^*, h^*\} \sim p_\theta(v, h).$$

Similarly, for biases:

$$\frac{\partial \mathcal{L}}{\partial a_i} = \langle v_i \rangle_{\text{data}} - \langle v_i \rangle_{\text{model}}, \quad \frac{\partial \mathcal{L}}{\partial b_j} = \langle h_j \rangle_{\text{data}} - \langle h_j \rangle_{\text{model}}.$$

Gibbs Sampling for RBMs

Block Gibbs sampling alternates:

- 1 Sample $h \sim p_\theta(h \mid v)$: draw each h_j independently from $\text{Bernoulli}(\sigma(b_j + W_{:j}^\top v))$.
- 2 Sample $v' \sim p_\theta(v \mid h)$: draw each v_i independently from $\text{Bernoulli}(\sigma(a_i + W_{i:} h))$.

Starting from any $v^{(0)}$, iterating k steps gives samples from $p_k(\cdot) = p_\theta(\cdot \mid T^k)$ where T is the Gibbs transition.

Convergence: as $k \rightarrow \infty$, $p_k \rightarrow p_\theta$ (ergodic Markov chain). CD- k uses the k -step approximation in lieu of the true equilibrium.

The Bottleneck and the CD Idea

The exact model expectation

$$\langle v_i h_j \rangle_{\text{model}}$$

requires Gibbs chains run to equilibrium — expensive or impractical.

Contrastive divergence (Hinton 2002) replaces equilibrium sampling by *short chains initialised at data*:

$$v^{(0)} \sim p_{\text{data}} \rightarrow h^{(0)} \sim p(h|v^{(0)}) \rightarrow v^{(1)} \sim p(v|h^{(0)}) \rightarrow \dots \rightarrow v^{(k)}.$$

The distribution of $v^{(k)}$ is denoted p_k . CD- k uses

$$\langle v_i h_j \rangle_{\text{model}} \approx \langle v_i h_j \rangle_k \implies \Delta W_{ij}^{\text{CD}-k} \propto \langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_k.$$

Why “Contrastive Divergence”? — The Objective

Define the **contrastive divergence objective**:

$$\text{CD}_k(\theta) = D_{\text{KL}}(p_0 \| p_\theta) - D_{\text{KL}}(p_k \| p_\theta),$$

where $p_0 = p_{\text{data}}$ and $p_k = p_0 T_\theta^k$ (T_θ = Gibbs kernel leaving p_θ invariant).

Interpretation.

- The first term pushes p_θ toward the data distribution.
- The second term *undoes* some of that push, using the short-run reconstruction distribution p_k .
- The *contrast* is between the data divergence and the k -step reconstruction divergence.

As $k \rightarrow \infty$, $p_k \rightarrow p_\theta$, so $\text{CD}_k \rightarrow D_{\text{KL}}(p_0 \| p_\theta)$ (exact ML).

Gradient of the First KL Term

Since $p_0 = p_{\text{data}}$ does not depend on θ :

$$\begin{aligned}\nabla_{\theta} D_{\text{KL}}(p_0 \| p_{\theta}) &= -\nabla_{\theta} \mathbb{E}_{p_0}[\log p_{\theta}(x)] \\ &= -\mathbb{E}_{p_0}[\nabla_{\theta} \log p_{\theta}(x)] \\ &= \mathbb{E}_{p_0}[\nabla_{\theta} E_{\theta}(x)] - \mathbb{E}_{p_{\theta}}[\nabla_{\theta} E_{\theta}(x)] \\ &= \langle \nabla_{\theta} E_{\theta} \rangle_{p_0} - \langle \nabla_{\theta} E_{\theta} \rangle_{p_{\theta}}.\end{aligned}$$

This is the *exact* ML gradient: positive phase minus negative phase.

Gradient of the Second KL Term

$$D_{\text{KL}}(p_k \| p_\theta) = \sum_x p_k(x) \log \frac{p_k(x)}{p_\theta(x)}.$$

The full gradient has two contributions:

- 1 **Explicit:** derivative through p_θ (treating p_k as fixed).
- 2 **Implicit:** derivative through $p_k = p_0 T_\theta^k$, since T_θ depends on θ .

The **CD approximation** neglects the implicit term:

$$\nabla_\theta p_k \approx 0.$$

Under this approximation:

$$\nabla_\theta D_{\text{KL}}(p_k \| p_\theta) \approx \langle \nabla_\theta E_\theta \rangle_{p_k} - \langle \nabla_\theta E_\theta \rangle_{p_\theta}.$$

The CD Gradient: Cancellation

$$\begin{aligned}\nabla_{\theta} \text{CD}_k &\approx \nabla_{\theta} D_{\text{KL}}(p_0 \| p_{\theta}) - \nabla_{\theta} D_{\text{KL}}(p_k \| p_{\theta}) \\ &= \left[\langle \nabla_{\theta} E_{\theta} \rangle_{p_0} - \langle \nabla_{\theta} E_{\theta} \rangle_{p_{\theta}} \right] - \left[\langle \nabla_{\theta} E_{\theta} \rangle_{p_k} - \langle \nabla_{\theta} E_{\theta} \rangle_{p_{\theta}} \right].\end{aligned}$$

The model expectations $\langle \nabla_{\theta} E_{\theta} \rangle_{p_{\theta}}$ cancel:

$$\nabla_{\theta} \text{CD}_k \approx \langle \nabla_{\theta} E_{\theta} \rangle_{p_0} - \langle \nabla_{\theta} E_{\theta} \rangle_{p_k}.$$

This is the final CD update rule: data statistics minus k -step reconstruction statistics. The intractable partition-function gradient has been eliminated.

Concrete RBM CD-1 Update

For CD-1 on an RBM, one Gibbs step gives $v^{(1)}$ from $v^{(0)} \sim p_{\text{data}}$:

$$h_j^{(0)} \sim \text{Bernoulli}(\sigma(b_j + W_{:j}^\top v^{(0)})),$$

$$v_i^{(1)} \sim \text{Bernoulli}(\sigma(a_i + W_{i:} h^{(0)})),$$

$$h_j^{(1)} \sim \text{Bernoulli}(\sigma(b_j + W_{:j}^\top v^{(1)})).$$

The CD-1 weight update is then:

$$\Delta W_{ij} = \eta \left(v_i^{(0)} \mathbb{E}[h_j^{(0)} \mid v^{(0)}] - v_i^{(1)} \mathbb{E}[h_j^{(1)} \mid v^{(1)}] \right),$$

$$\Delta a_i = \eta (v_i^{(0)} - v_i^{(1)}), \quad \Delta b_j = \eta (\mathbb{E}[h_j^{(0)} \mid v^{(0)}] - \mathbb{E}[h_j^{(1)} \mid v^{(1)}]).$$

Using the *mean-field* hidden activations (rather than samples) reduces variance.

CD- k is *not* an unbiased estimator of the ML gradient.

The bias. Let T_θ be the Gibbs transition kernel. The true gradient of $D_{\text{KL}}(p_k \| p_\theta)$ is

$$\nabla_\theta D_{\text{KL}}(p_k \| p_\theta) = \langle \nabla_\theta E_\theta \rangle_{p_k} - \langle \nabla_\theta E_\theta \rangle_{p_\theta} + \underbrace{\sum_x (\nabla_\theta p_k(x)) \log \frac{p_k(x)}{p_\theta(x)}}_{\text{neglected implicit term}}.$$

Neglecting the implicit term introduces a bias that vanishes as $k \rightarrow \infty$.

Consequence. CD- k may not converge to the true ML solution and can exhibit slow drift or oscillation during training.

Persistent Contrastive Divergence (PCD)

Key idea (Tieleman 2008): maintain a persistent set of fantasy particles $\{v^{(n)}\}_{n=1}^N$ that are *not* reinitialised from data.

PCD update:

- 1 Run one Gibbs step from each fantasy particle: $v^{(n)} \rightarrow v^{(n)'}$.
- 2 Compute: $\Delta W_{ij} = \eta(\langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_{\text{fantasy}})$.
- 3 Update the fantasy particles: $v^{(n)} \leftarrow v^{(n)'}$.

Why better? Fantasy particles mix across modes between updates; they approximate the true model distribution more faithfully than CD- k short chains.

		CD- k	PCD
CD vs PCD:	Init. of chain	data	persistent fantasy
	Bias	larger	smaller
	Mixing	local	global

Stochastic Maximum Likelihood (SML)

SML is essentially PCD with a rigorous motivation.

At step t , the Markov chain is at $v^{(t)}$, which approximates a sample from $p_{\theta^{(t)}}$. After a small parameter update $\theta^{(t)} \rightarrow \theta^{(t+1)}$, $v^{(t)}$ is still close to a sample from $p_{\theta^{(t+1)}}$ provided the update is small.

Gradient approximation:

$$\nabla_{\theta} \mathcal{L}(\theta) \approx \langle \nabla_{\theta} E_{\theta} \rangle_{\text{data}} - \nabla_{\theta} E_{\theta}(v^{(t)}).$$

The approximation improves as the learning rate decreases, and under mild conditions SML is asymptotically unbiased.

Natural Gradient and Fisher Information

Standard gradient ascent on $\mathcal{L}(\theta)$ is geometry-unaware.

The **natural gradient** (Amari 1998) uses the *Fisher information matrix* $F(\theta)$:

$$F_{ab}(\theta) = \mathbb{E}_{p_\theta} \left[\frac{\partial \log p_\theta}{\partial \theta_a} \frac{\partial \log p_\theta}{\partial \theta_b} \right],$$

and updates

$$\theta \leftarrow \theta + \eta F(\theta)^{-1} \nabla_\theta \mathcal{L}(\theta).$$

For EBM:

$$F_{ab}(\theta) = \text{Cov}_{p_\theta} [\nabla_{\theta_a} E_\theta, \nabla_{\theta_b} E_\theta].$$

Computing F^{-1} is expensive, but the natural gradient converges faster and is invariant to reparameterisation. Approximate versions (e.g., K-FAC) are used in practice.

Energy Landscape Picture

For $p_{\theta}(x) \propto e^{-E_{\theta}(x)}$, learning reshapes the energy surface:

ML Lower E on data; raise E on model samples.

CD- k Lower E on data; raise E on k -step reconstructions.

The CD update is *local*: it adjusts the energy landscape only in the neighbourhood of data points, as defined by the mixing distance of k Gibbs steps.

This locality is both a strength (cheap) and a weakness (misses distant modes).

Relation to Other EBM Training Methods

Method	Negative samples	Notes
CD- k	short Gibbs chains	biased, cheap
PCD / SML	persistent chains	less biased
Score matching	analytic	no sampling needed
Noise-contrastive est.	fixed noise dist.	binary classifier
Diffusion-based	learned reverse SDE	SOTA generative models

CD uses the model's own dynamics to generate negative samples. Unlike score matching, it does not require second-order derivatives of E_θ . Unlike NCE, it does not need an explicit noise distribution.

Summary: Classical CD

For a classical EBM $p_\theta(x) = e^{-E_\theta(x)}/Z_\theta$, ML maximisation gives the intractable gradient

$$\nabla_\theta \mathcal{L} = -\langle \nabla_\theta E_\theta \rangle_{\text{data}} + \langle \nabla_\theta E_\theta \rangle_{\text{model}}.$$

CD- k approximates this by replacing equilibrium model expectations with k -step Gibbs reconstructions:

$$\nabla_\theta \text{CD}_k \approx \langle \nabla_\theta E_\theta \rangle_{p_0} - \langle \nabla_\theta E_\theta \rangle_{p_k}.$$

This approximation is **biased** (neglects $\nabla_\theta p_k$) but **efficient** (no partition function needed).

Summary: Why the Name?

The CD objective is

$$\text{CD}_k(\theta) = \underbrace{D_{\text{KL}}(p_0 \| p_\theta)}_{\text{data divergence}} - \underbrace{D_{\text{KL}}(p_k \| p_\theta)}_{\text{reconstruction divergence}} .$$

Contrastive Compare (contrast) data statistics with short-run reconstruction statistics.

Divergence The comparison is formalised through KL divergences.

As k increases the approximation improves:

$$k = 0 : \quad \text{CD}_0 = 0; \quad k \rightarrow \infty : \quad \text{CD}_k \rightarrow D_{\text{KL}}(p_0 \| p_\theta) \text{ (= exact ML).}$$